

Q1 - 24 January - Shift 1

The compound statement

$$(\sim(P \wedge Q)) \vee ((\sim P) \wedge Q) \Rightarrow ((\sim P) \wedge (\sim Q))$$

is equivalent to

(1) $((\sim P) \vee Q) \wedge ((\sim Q) \vee P)$

(2) $(\sim Q) \vee P$

(3) $((\sim P) \vee Q) \wedge (\sim Q)$

(4) $(\sim P) \vee Q$

Space for your notes:

Q2 - 24 January - Shift 2

Let p and q be two statements.

Then $\sim(p \wedge (p \Rightarrow \sim q))$ is equivalent to

(1) $p \vee (p \wedge (\sim q))$

(2) $p \vee ((\sim p) \wedge q)$

(3) $(\sim p) \vee q$

(4) $p \vee (p \wedge q)$

Space for your notes:

Q3 - 25 January - Shift 1

The statement $(p \wedge (\sim q)) \Rightarrow (p \Rightarrow (\sim q))$ is

(1) equivalent to $(\sim p) \vee (\sim q)$

(2) a tautology

(3) equivalent to $p \vee q$

(4) a contradiction

Space for your notes:

Q4 - 25 January - Shift 2

Let $\Delta, \nabla \in \{\wedge, \vee\}$ be such that $(p \rightarrow q) \Delta (p \vee q)$

is a tautology. Then

- (1) $\Delta = \wedge, \nabla = \vee$ (2) $\Delta = \vee, \nabla = \wedge$
 (3) $\Delta = \vee, \nabla = \vee$ (4) $\Delta = \wedge, \nabla = \wedge$

Space for your notes:

Q5 - 29 January - Shift 1

If p, q and r are three propositions, then which of the following combination of truth values of p, q and r makes the logical expression

$\{(p \vee q) \wedge ((\sim p) \vee r)\} \rightarrow ((\sim q) \vee r)$ false ?

- (1) $p = T, q = F, r = T$ (2) $p = T, q = T, r = F$
 (3) $p = F, q = T, r = F$ (4) $p = T, q = F, r = F$

Space for your notes:

Q6 - 29 January - Shift 2

The statement $B \Rightarrow ((\sim A) \vee B)$ is equivalent to

- (1) $B \Rightarrow (A \Rightarrow B)$
 (2) $A \Rightarrow (A \Leftrightarrow B)$
 (3) $A \Rightarrow ((\sim A) \Rightarrow B)$
 (4) $B \Rightarrow ((\sim A) \Rightarrow B)$

Space for your notes:

Q7 - 30 January - Shift 1

Among the statements:

$$(S1) \quad ((p \vee q) \Rightarrow r) \Leftrightarrow (p \Rightarrow r)$$

$$(S2) \quad ((p \vee q) \Rightarrow r) \Leftrightarrow ((p \Rightarrow r) \vee (q \Rightarrow r))$$

- (1) Only (S1) is a tautology
- (2) Neither (S1) nor (S2) is a tautology
- (3) Only (S2) is a tautology
- (4) Both (S1) and (S2) are tautologies

Space for your notes:

Q8 - 30 January - Shift 2

Consider the following statements:

P : I have fever

Q : I will not take medicine

R : I will take rest

The statement "If I have fever, then I will take medicine and I will take rest" is equivalent to:

$$(1) \quad ((\sim P) \vee \sim Q) \wedge ((\sim P) \vee R)$$

$$(2) \quad ((\sim P) \vee \sim Q) \wedge ((\sim P) \vee \sim R)$$

$$(3) \quad (P \vee Q) \wedge ((\sim P) \vee R)$$

$$(4) \quad (P \vee \sim Q) \wedge (P \vee \sim R)$$

Space for your notes:

Q9 - 31 January - Shift 1

(S1) $(p \Rightarrow q) \vee (p \wedge (\sim q))$ is a tautology

Space for your notes:

(S2) $((\sim p) \Rightarrow (\sim q)) \wedge ((\sim p) \vee q)$ is a

Contradiction. Then

- (1) only (S2) is correct
- (2) both (S1) and (S2) are correct
- (3) both (S1) and (S2) are wrong
- (4) only (S1) is correct

Q10 - 31 January - Shift 2

The number of values of $r \in \{p, q, \sim p, \sim q\}$ for which $((p \wedge q) \Rightarrow (r \vee q)) \wedge ((p \wedge r) \Rightarrow q)$ is a tautology, is:

Space for your notes:

- (1) 3
- (2) 2
- (3) 1
- (4) 4

Q11 - 01 February - Shift 1

The negation of the expression $q \vee ((\sim q) \wedge p)$ is equivalent to

Space for your notes:

- (1) $(\sim p) \wedge (\sim q)$
- (2) $p \wedge (\sim q)$
- (3) $(\sim p) \vee (\sim q)$
- (4) $(\sim p) \vee q$

Q12 - 01 February - Shift 2

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Which of the following statements is a tautology ?

Space for your notes:

(1) $p \rightarrow (p \wedge (p \rightarrow q))$

(2) $(p \wedge q) \rightarrow (\sim(p) \rightarrow q)$

(3) $(p \wedge (p \rightarrow q)) \rightarrow \sim q$

(4) $p \vee (p \wedge q)$

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (1)

Q2 (3)

Q3 (2)

Q4 (3)

Q5 (3)

Q6 (2)

Q7 (2)

Q8 (1)

Q9 (2)

Q10 (2)

Q11 (1)

Q12 (2)

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Q1 (1)

Let $r = (\sim(P \wedge Q)) \vee ((\sim P) \wedge Q)$; $s = ((\sim P) \wedge (\sim Q))$

P	Q	$\sim(P \wedge Q)$	$(\sim P) \wedge Q$	r	s	$r \rightarrow s$
T	T	F	F	F	F	T
T	F	T	F	T	F	F
F	T	T	T	T	F	F
F	F	T	F	T	T	T

Option (A) : $((\sim P) \vee Q) \wedge ((\sim Q) \vee P)$

is equivalent to (not of only P) $\wedge$ (not of only Q)
 = (Both P, Q) and (neither P nor Q)

Q2 (3)

$$\sim(p \wedge (p \rightarrow \sim q))$$

$$\equiv \sim p \vee \sim(\sim p \vee \sim q)$$

$$\equiv \sim p \vee (p \wedge q)$$

$$\equiv (\sim p \vee p) \wedge (\sim p \vee q)$$

$$\equiv t \wedge (\sim p \vee q)$$

$$\equiv \sim p \vee q$$

Q3 (2)

$$\begin{aligned}(p \wedge \sim q) &\rightarrow (p \rightarrow \sim q) \\ &\equiv (\sim (p \wedge \sim q)) \vee (\sim p \vee \sim q) \\ &\equiv (\sim p \vee q) \vee (\sim p \vee \sim q) \\ &\equiv \sim p \vee t \equiv t\end{aligned}$$

Q4 (3)

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Given $(p \rightarrow q) \Delta (p \nabla q)$

Option I $\Delta = \wedge, \nabla = \vee$

p	q	$(p \rightarrow q)$	$(p \vee q)$	$(p \rightarrow q) \wedge (p \vee q)$
T	T	T	T	T
T	F	F	T	F
F	T	T	T	T
F	F	T	F	F

Option 2 $\Delta = \vee, \nabla = \wedge$

p	q	$(p \rightarrow q)$	$(p \wedge q)$	$(p \rightarrow q) \vee (p \wedge q)$
T	T	T	T	T
T	F	F	F	F
F	T	T	F	T
F	F	T	F	T

Option 3 $\Delta = \vee, \nabla = \vee$

p	q	$(p \rightarrow q)$	$(p \vee q)$	$(p \rightarrow q) \vee (p \wedge q)$
T	T	T	T	T
T	F	F	T	T
F	T	T	T	T
F	F	T	F	T

Hence, it is tautology.

Option 4 $\Delta = \wedge, \nabla = \wedge$

p	q	$(p \rightarrow q)$	$(p \wedge q)$	$(p \rightarrow q) \wedge (p \wedge q)$
T	T	T	T	T
T	F	F	F	F
F	T	T	F	F
F	F	T	F	F

Q5 (3)

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	p	q	r	$(p \vee q) \wedge ((\sim p) \vee r)$	$\sim q \vee r$
(1)	T	F	T	T	T
(2)	T	T	F	F	F
(3)	F	T	F	T	F
(4)	T	F	F	F	T

Option (3) $(p \vee q) \wedge (\sim q \vee r) \rightarrow (\sim p \vee r)$ will be False.

Q6 (2)

A	B	$\sim A$	$\sim A \vee B$	$B \Rightarrow ((\sim A) \vee B)$
T	T	F	T	T
T	F	F	F	T
F	T	T	T	T
F	F	T	T	T

$A \Rightarrow B$	$\sim A \Rightarrow B$	$B \Rightarrow (A \Rightarrow B)$	$A \Rightarrow ((\sim A) \Rightarrow B)$	$B \Rightarrow ((\sim A) \Rightarrow B)$
T	T	T	T	T
F	T	T	T	T
T	T	T	T	T
T	F	T	T	T

Q7 (2)

$$S_1 \equiv ((p \vee q) \Rightarrow r) \Leftrightarrow (p \Rightarrow r)$$

p	q	r	$p \vee q$	$(p \vee q) \Rightarrow r$	$p \Rightarrow r$	$((p \vee q) \Rightarrow r) \Leftrightarrow (p \Rightarrow r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	T
T	F	T	T	T	T	T
T	F	F	T	F	F	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	T
F	F	F	F	T	T	T

$$S_2 \equiv (p \vee q) \Rightarrow r \Leftrightarrow ((p \Rightarrow r) \vee (q \Rightarrow r))$$

p	q	r	$(p \vee q) \Rightarrow r$	$p \Rightarrow r$	$q \Rightarrow r$	$(p \Rightarrow r) \vee (q \Rightarrow r)$	S_2
T	T	T	T	T	T	T	T
T	T	F	F	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	F	F	T	T	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

$$S_2 \rightarrow \text{not a tautology}$$

$$S_2 \rightarrow \text{not a tautology}$$

$$S_2 \rightarrow \text{not a tautology}$$

$$S_2 \rightarrow \text{not a tautology}$$

$$S_2 \rightarrow \text{not a tautology}$$

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$$S_2 \rightarrow \text{not a tautology}$$

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p	q	$p \Rightarrow q$	$\sim q$	$p \wedge \sim q$	$(p \Rightarrow q) \vee (p \wedge \sim q)$
T	T	T	F	F	T
T	F	F	T	T	T
F	T	T	F	F	T
F	F	T	T	F	T

$\sim p$	$\sim q$	$\sim p \Rightarrow \sim q$	$\sim p \vee q$	$((\sim p) \Rightarrow (\sim q)) \wedge (\sim p) \vee q$
F	F	T	T	T
F	T	T	F	F
T	F	F	T	F
T	T	T	T	T

Q10 (2)

$$((p \wedge q) \Rightarrow (r \vee q)) \wedge ((p \wedge r) \Rightarrow q)$$

We know, $p \Rightarrow q$ is equivalent to

$$\begin{aligned} & \sim p \vee q \\ & (\sim(p \wedge q) \vee (r \vee q)) \wedge (\sim(p \wedge r) \vee q) \\ & \Rightarrow (\sim p \vee \sim q \vee r \vee q) \wedge (\sim p \vee \sim r \vee q) \\ & \Rightarrow (\sim p \vee r \vee t) \wedge (\sim p \vee \sim r \vee q) \\ & \Rightarrow (t) \wedge (\sim p \vee \sim r \vee q) \end{aligned}$$

For this to be tautology, $(\sim p \vee \sim r \vee q)$ must be always true which follows for $r = \sim p$ or $r = q$.

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Q11 (1)

$$\begin{aligned}
 & \sim (q \vee ((\sim q) \wedge p)) \\
 &= \sim q \wedge \sim ((\sim q) \wedge p) \\
 &= \sim q \wedge (q \vee \sim p) \\
 &= (\sim q \wedge q) \vee (\sim q \wedge \sim p) \\
 &= (\sim q \wedge \sim p)
 \end{aligned}$$

Q12 (2)

$$\begin{aligned}
 & \text{(i) } p \rightarrow (p \wedge (p \rightarrow q)) \\
 & (\sim p) \vee (p \wedge (\sim p \vee q)) \\
 & (\sim p) \vee (p \wedge q) \\
 & \sim p \vee (p \wedge q) = (\sim p \vee p) \wedge (\sim p \vee q) \\
 &= \sim p \vee q \\
 & \text{(ii) } (p \wedge q) \rightarrow (\sim p \rightarrow q) \\
 & \sim(p \wedge q) \vee (p \vee q) = t \\
 & \{a, b, d\} \vee \{a, b, c\} = V \\
 & \text{Tautology} \\
 & \text{(iii) } (p \wedge (p \rightarrow q)) \rightarrow \sim q \\
 & \sim(p \wedge (\sim p \vee q)) \vee \sim q = \sim(p \wedge q) \vee \sim q = \sim p \vee \sim q \\
 & \text{Not tautology} \\
 & \text{(iv) } p \vee (p \wedge q) = p \\
 & \text{Not tautology.}
 \end{aligned}$$